

India Inc.'s roll

There's been a lot of India-related news in the international job and outsourcing front. The American state of Colorado voted to postpone a bill that prevented jobs from going to offshore workers, especially for state contracts. US businesses find outsourcing to India a cheaper deal than getting work done locally. Moreover, India's big pool of talented and highly skilled workers works for far less than their American counterparts.

In related news, the US government has exhausted their current quota of H1-B visas for the current fiscal year, with no visa applications being accepted until April 2004, and none being issued till October of this year. The H1-B visa programme attracts large numbers of skilled Indian technologists, and has been very controversial in the US.

In yet another development, a US Congressional panel raised concerns on the apparent misuse of L-1 visas that allow companies to get in skilled workers in executive and managerial positions, with no limits on the numbers issued, or a required pay limit. Witnesses allege large-scale cases of fraud to get in temporary workers into the US, and the associated loss of jobs for locals.

snapshot

The value of **Time Warner Inc.** is now **US\$ 42 billion** lesser than their value before merger with **AOL**

Source: Fortune

Smile, you're on CCD

Take a digital camera, and you have found yourself a Charge Coupled Device (CCD); move lower down the digital chain to that camera-phone you have been eyeing, and a Complementary Metal-Oxide Semiconductor (CMOS) stares back. Both the technologies are meant to capture images, but they differ in cost and quality offered.

A CCD is ideal for high-quality captures, and has good low-light sensitivity. However, it is expensive to produce and a drain on the batteries. A CMOS is cheap to make, and lighter on the batteries. Its image quality however, leaves much to be desired. Can't have it all... or can you?

Japanese major Matsushita has taken the first step toward creating Maicovicon, a

hybrid sensor that has the advantages of both the chips. *EETimes* reported that Maicovicon consumes less power than a CCD, and offers higher image quality than a CMOS sensor. It will be available in



1.3 megapixel and 2 megapixel versions with a pixel size of 2.8 mm square. Photosensitivity is about three times that of CMOS sensors and almost equal to that of CCD sensors.

This hybrid should herald a new wave of low-powered devices sporting affordable and better-quality cameras.

tomorrow's technology

Siliconconnect

Intel researchers have developed a silicon-based optical modulator operating at 1 GHz—an increase of over 50 times the previous research record of about 20 MHz. The device incorporates a transistor-like structure to encode data onto a wavelength of light. The modulator splits the incoming light beam into two. These are then phase-shifted to change the amplitude of the resulting, recombined beam. The result is the ability to change light from bright to dark and thus, encode data.

This allows for faster connections between PCs, servers,

and manufacturing of faster buses and interconnects inside a PC. While CPUs are projected to meet future bandwidth demands, the bandwidth of the interconnects needs to be increased to prevent bottlenecks. Photonics (using photons as data carriers instead of electrons) offers greater bandwidth than copper networks, and carries multiple signals simultaneously without interference. However, commercial photonic devices are difficult to assemble and expensive. This high cost limits their use to special applications such as telecommunications and wide-area networks.

quoteworthy

"I may have invented [Control]+[Alt]+[Delete], but Bill Gates made it really famous"

David Bradley, now retired IBM employee who did pioneering work on the original IBM PC. Bradley was speaking to a gathering on the 20th anniversary of the PC.

"I never expected it to be precise. It turned out to be much more precise than I ever imagined"

Gordon Moore, Co-founder of Intel, who formulated Moore's Law, which states that the number of transistors on a chip would double every year, and later, 20 months; Intel executive, David House, later accurately re-calculated it as 18 months.

"There's no reason to wait for any other x86, 64-bit technology, because we have it now, we're shipping it now"

John Robinson, AMD's country manager for Australia and New Zealand, talking to iNews Australia. Robinson claimed to be baffled after Intel announced its plans of shipping new Xeon processors with 64-bit extensions on the existing 32-bit core by the next quarter. AMD has already got a 64-bit processor for the desktop, the Athlon 64.

■ Intel committed to offering full-scale Linux driver support for their hardware using the same driver update and release cycle as that for Windows